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(54) **VALVE ARRANGEMENT FOR SPLIT-FLOW
CLOSE-COUPLED CATALYST**

(58) **Field of Classification Search**

CPC combination set(s) only.

See application file for complete search history.

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F01N 3/28 (2006.01)
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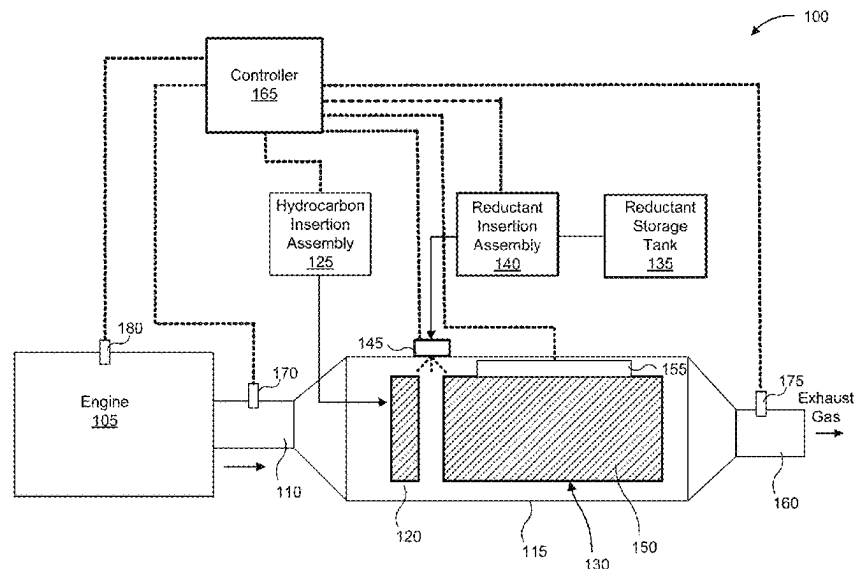
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(57) **ABSTRACT**

An aftertreatment system includes: a first exhaust gas path comprising a heater; a second exhaust gas path comprising a first decomposition chamber configured to receive reductant and a first selective catalytic reduction catalyst downstream of the first decomposition chamber; a combined exhaust gas path downstream of the first exhaust gas path and the second exhaust gas path, the combined exhaust gas path configured to receive exhaust gas from both the first exhaust gas path and the second exhaust gas path; a selector valve configured to divert the exhaust gas between the first exhaust gas path and the second exhaust gas path based on a temperature of the exhaust gas; and a controller programmed to control the selector valve.

20 Claims, 6 Drawing Sheets



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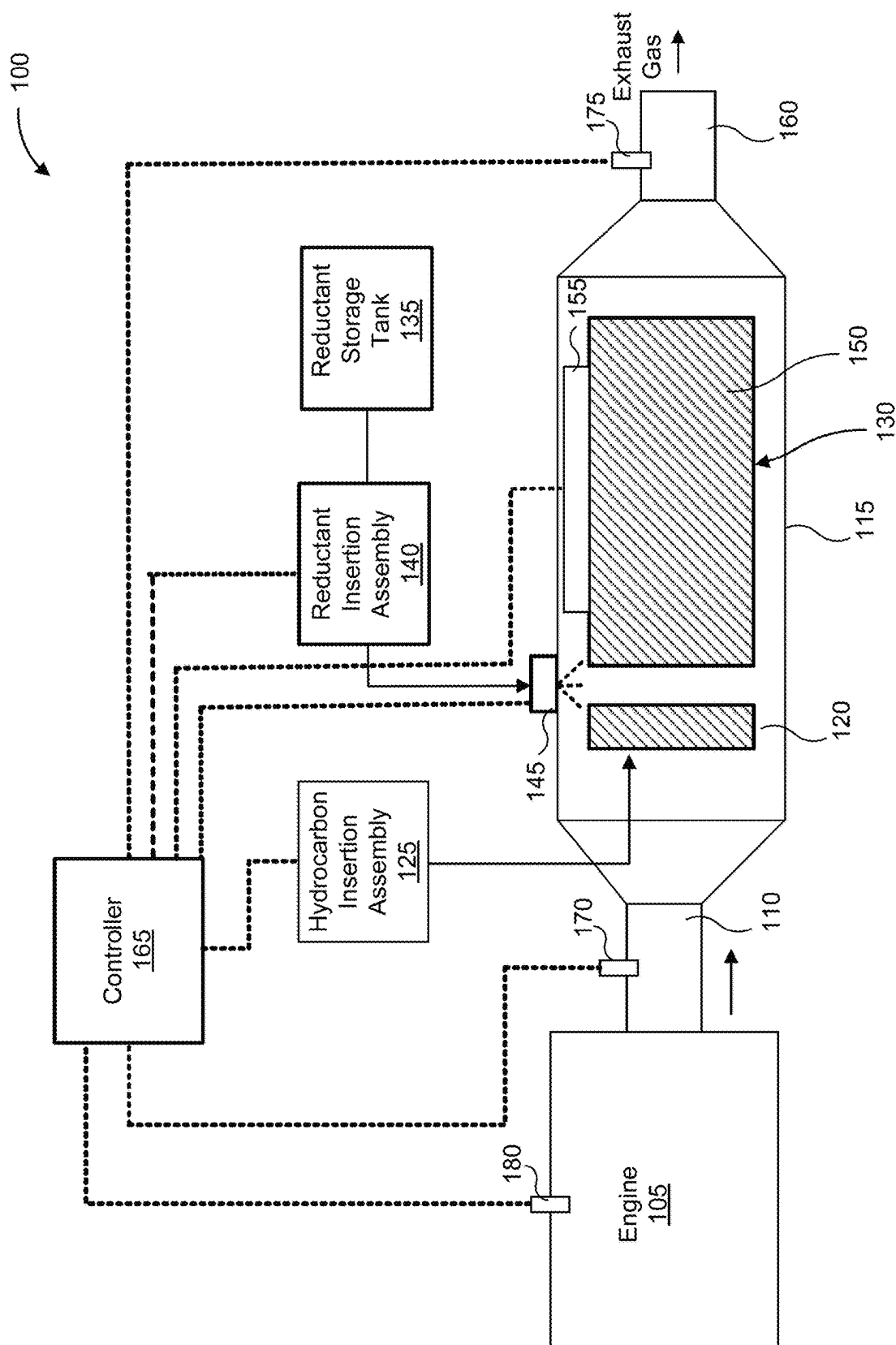


FIG. 1

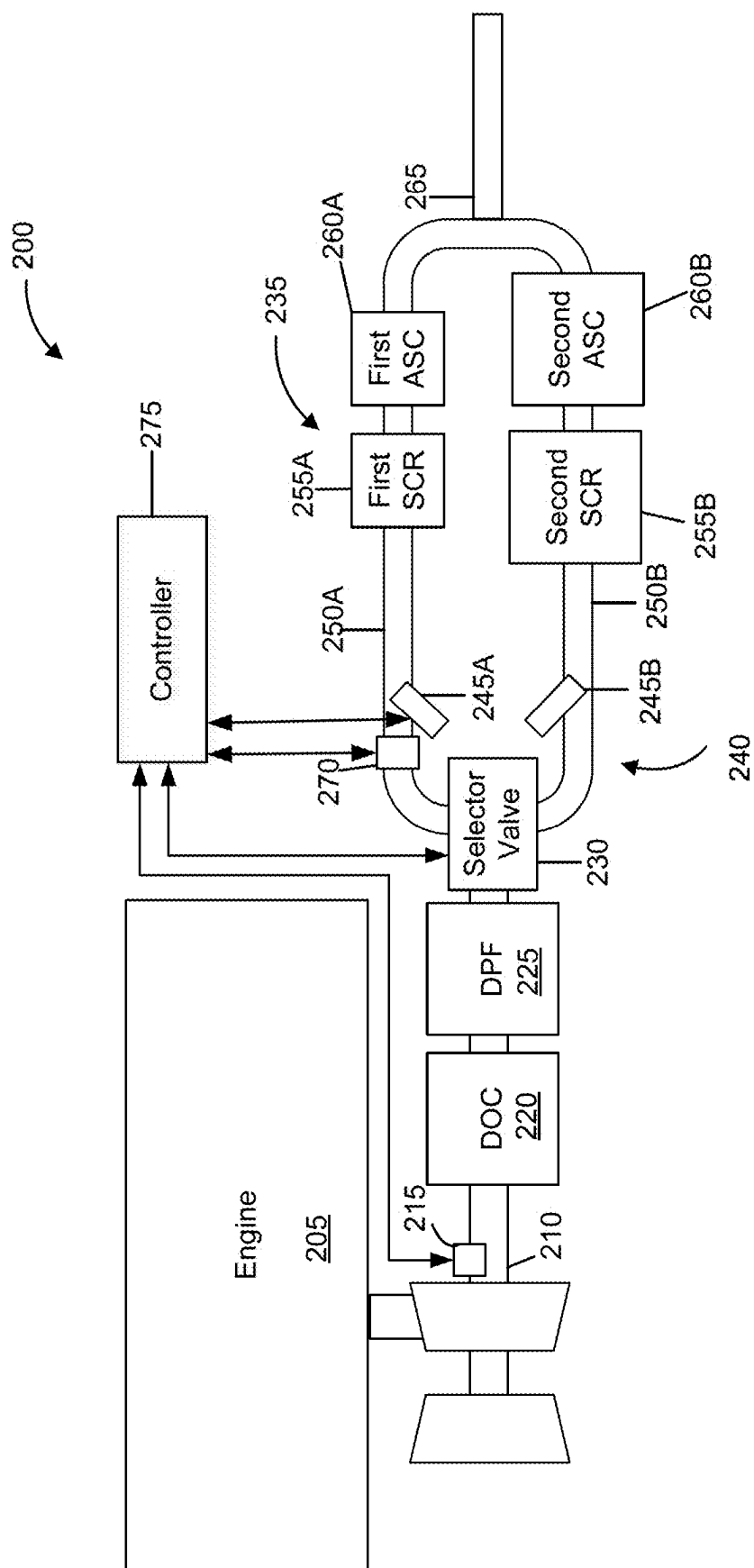


FIG. 2

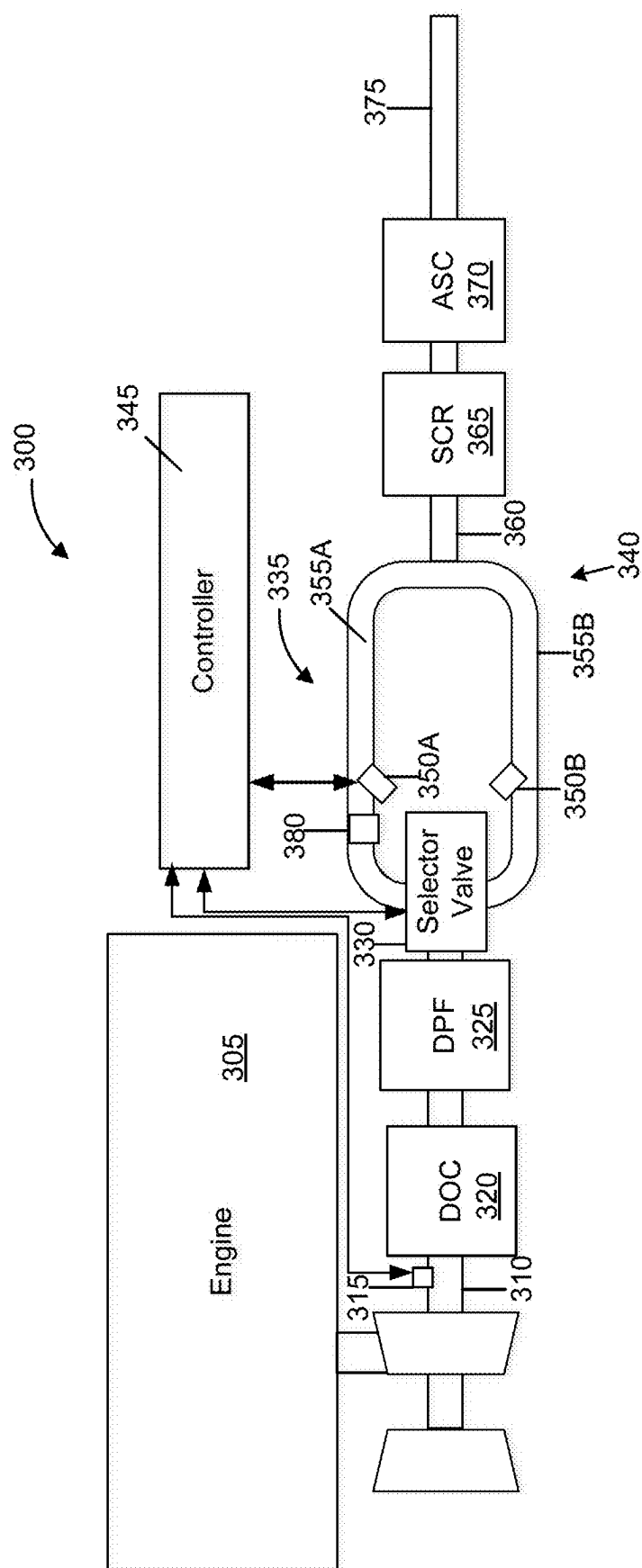


FIG. 3

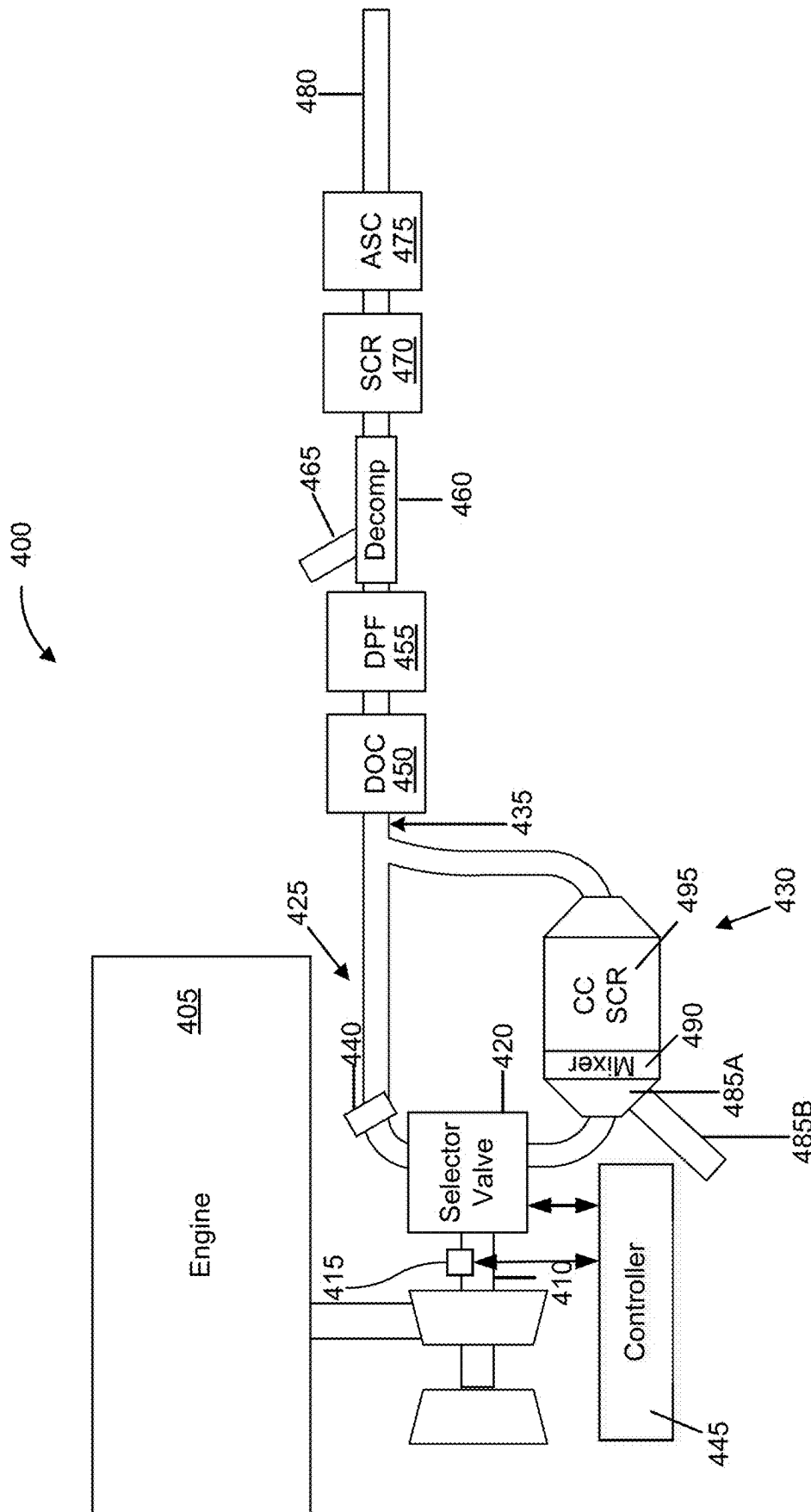


FIG. 4A

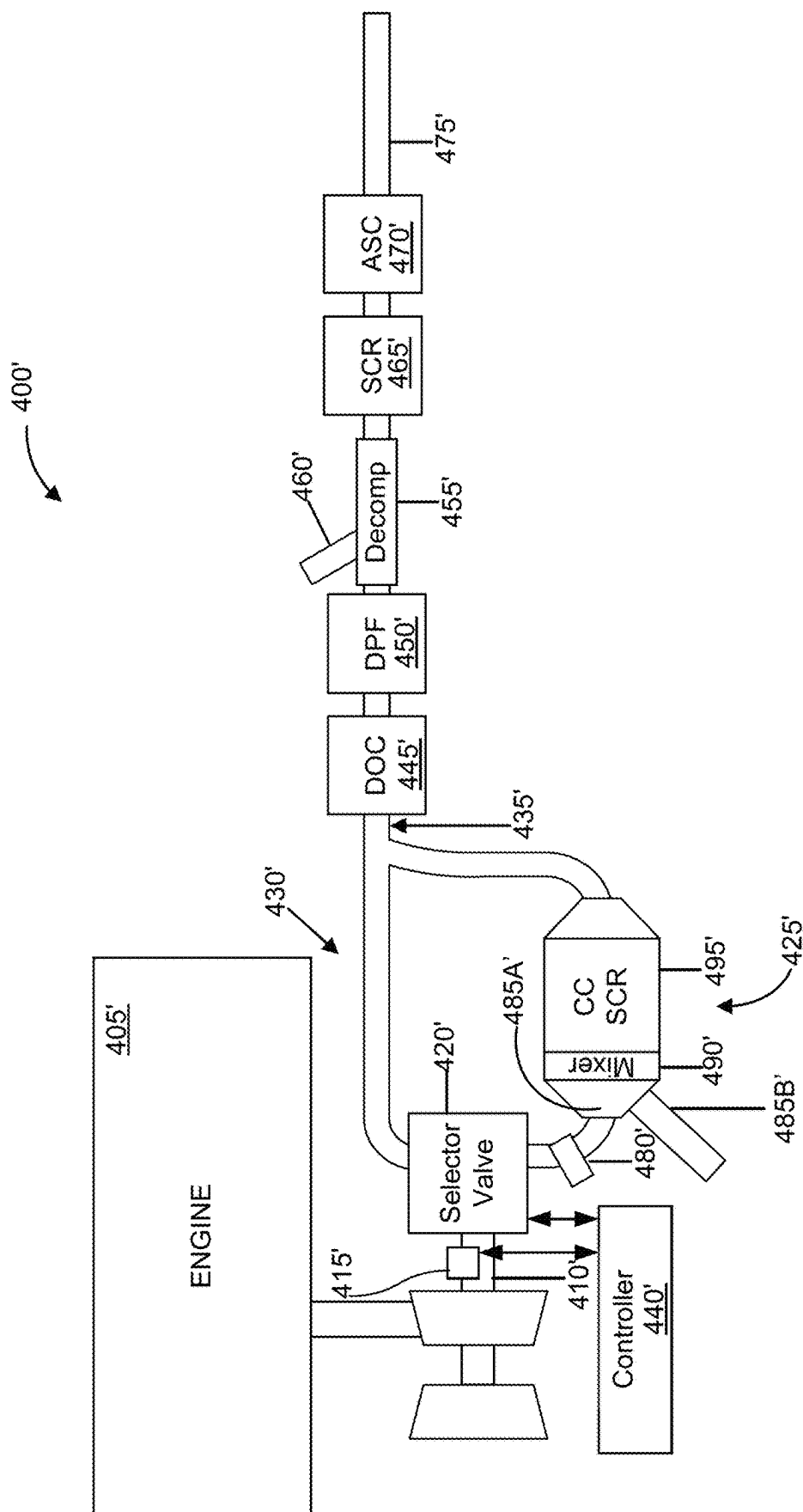


FIG. 4B

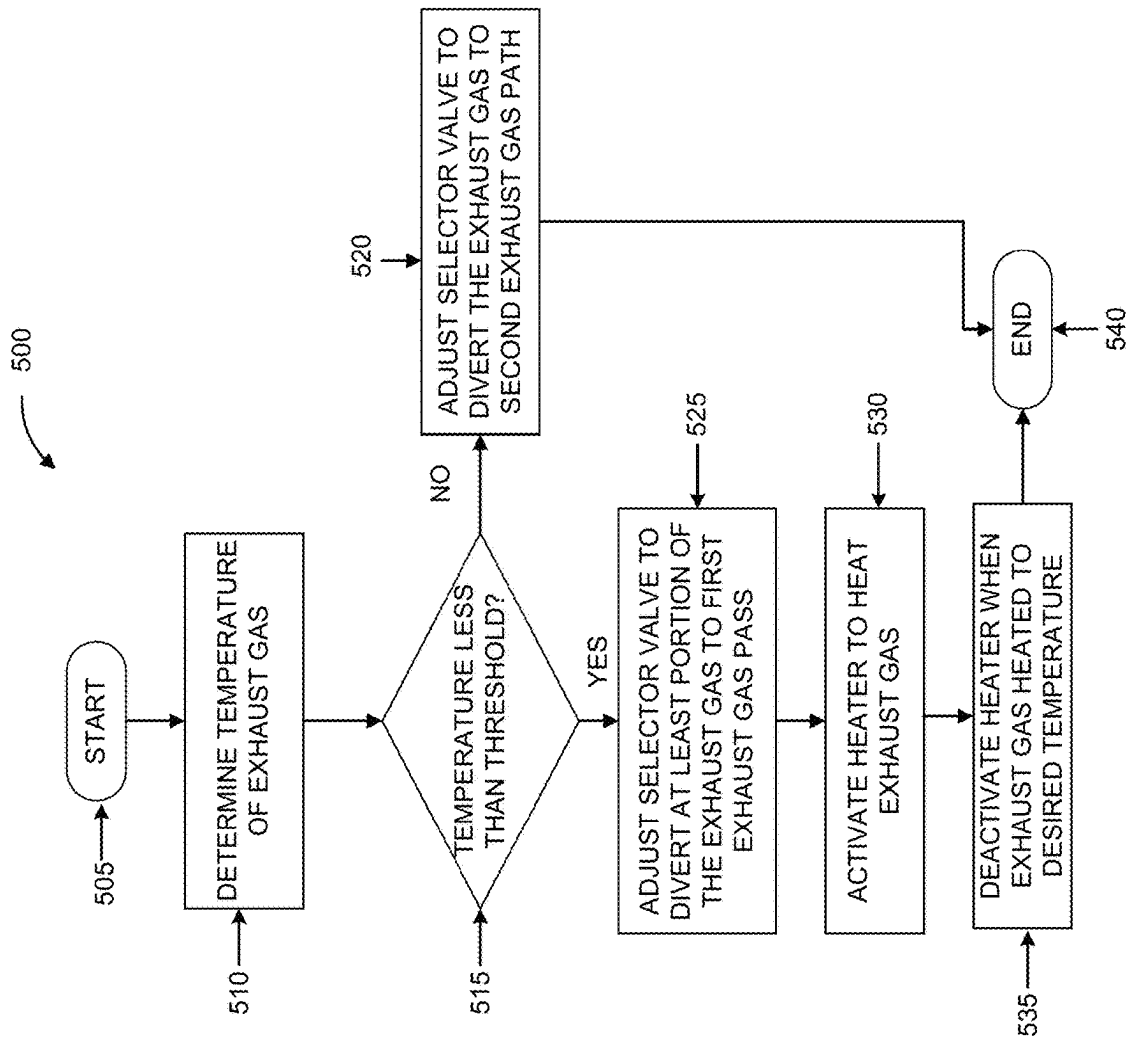


FIG. 5

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VALVE ARRANGEMENT FOR SPLIT-FLOW CLOSE-COUPLED CATALYST

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a continuation of U.S. patent application Ser. No. 17/609,681, filed Nov. 8, 2021, which is the U.S. National Stage of PCT Application No. PCT/US2019/031542, filed May 9, 2019. The entire contents of these applications are hereby incorporated by reference.

TECHNICAL FIELD

The present disclosure relates generally to aftertreatment systems for use with internal combustion engines.

BACKGROUND

An exhaust aftertreatment system is used to treat exhaust gas generated by an internal combustion engine. The exhaust aftertreatment system typically includes a selective catalytic reduction system that is formulated to reduce oxides of nitrogen in the exhaust gas in the presence of a catalyst and reductant. The exhaust aftertreatment system may also include one or more filters to remove debris and other particulates from the exhaust gas. By treating the exhaust gas using the exhaust aftertreatment system, the exhaust aftertreatment system reduces the levels of harmful emissions in the exhaust gas that would otherwise be emitted into the atmosphere. However, present day exhaust aftertreatment systems have limitations due to their configuration and the way they operate.

SUMMARY

In accordance with some aspects of the present disclosure, an aftertreatment system is disclosed. The aftertreatment system includes a first exhaust gas path, a second exhaust gas path, and a selector valve configured to divert exhaust gas between the first exhaust gas path and the second exhaust gas path based on a temperature of the exhaust gas. The aftertreatment system also includes a controller programmed to control the selector valve such that the selector valve diverts at least a portion of the exhaust gas to the first exhaust gas path when the temperature of the exhaust gas is equal to or less than a predetermined temperature threshold and the selector valve diverts the exhaust gas to the second exhaust gas path when the temperature of the exhaust gas is greater than the predetermined temperature threshold. The first exhaust gas path includes a heater configured to heat the exhaust gas received in the first exhaust gas path.

In accordance with some other aspects of the present disclosure, a method is disclosed. The method includes determining, by a controller associated with an aftertreatment system, a temperature of exhaust gas, comparing, by the controller, the temperature of the exhaust gas with a predetermined temperature threshold, and when the temperature of the exhaust gas is equal to or less than the predetermined temperature threshold, adjusting a selector valve to a first position so as to divert at least a portion of the exhaust gas to a first exhaust gas path, and heating the exhaust gas in the first exhaust gas path, and when the temperature of the exhaust gas is greater than the predetermined temperature threshold, adjusting the selector valve to

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a second position so as to divert at least a portion of the exhaust gas to a second exhaust gas path.

It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the subject matter disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several implementations in accordance with the disclosure and are therefore not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings.

FIG. 1 is an example block diagram of an aftertreatment system, in accordance with some embodiments of the present disclosure.

FIG. 2 is another example block diagram of the aftertreatment system showing a close coupled system in which exhaust gas is diverted to either a first exhaust gas path or a second exhaust gas path, in accordance with some embodiments of the present disclosure.

FIG. 3 is yet another example block diagram of the aftertreatment system showing another close coupled system in which the exhaust is diverted to either a first exhaust gas path or a second exhaust gas path, in accordance with some embodiments of the present disclosure.

FIGS. 4A and 4B are an example block diagram of the aftertreatment system showing close coupled systems in which the exhaust gas enters a combined exhaust gas path from either a first exhaust gas path or a second exhaust gas path, in accordance with some embodiments of the present disclosure.

FIG. 5 is an example flow diagram outlining operations for operating the aftertreatment systems of FIGS. 2-4B, in accordance with some embodiments of the present disclosure.

Reference is made to the accompanying drawings throughout the following detailed description. In the drawings, similar symbols typically identify similar components unless context dictates otherwise. The illustrative implementations described in the detailed description, drawings, and claims are not meant to be limiting. Other implementations may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein and illustrated in the figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and made part of this disclosure.

DETAILED DESCRIPTION

This application is directed to an aftertreatment system designed to treat exhaust gases emitted from an engine. The aftertreatment system may remove various types of undesirable constituents from the exhaust gas before releasing the treated exhaust gas into the atmosphere. The aftertreatment

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system may include a decomposition chamber, which receives a reductant that is converted into gaseous ammonia and mixed with the exhaust gas. The mixture of the exhaust gas and gaseous ammonia is diverted to a selective catalytic reduction ("SCR") system where the gaseous ammonia is catalyzed to reduce NOx in the exhaust gas.

The decomposition chamber and the SCR system are configured to operate optimally at certain temperatures (e.g., temperatures greater than 180° C.). At lower temperatures, such as those encountered during cold-start conditions, the reductant that is injected into the decomposition chamber is more susceptible to forming solid deposits on the walls of the decomposition chamber. The reductant that is deposited on the walls of the decomposition chamber is not converted into gaseous ammonia and does not mix with the exhaust gas. Without sufficient gaseous ammonia in the mixture, the reaction in the SCR system is impacted and desirable levels of NOx reduction are not achieved. Thus, increased amounts of reductant may be needed to achieve desirable levels of NOx reduction during low temperatures. Further, at lower temperatures, even if the decomposition chamber is operating efficiently, the SCR system may be unable to reach its desired levels of NOx reduction, thereby emitting greater amounts of NOx in the atmosphere and potentially violating certain emission regulations.

Further, the SCR system uses an SCR catalyst that needs to adsorb a sufficient amount of gaseous ammonia before reducing NOx in the exhaust gas. At certain temperatures (e.g., close to 300° C.), the adsorbed ammonia may be desorbed and lost to the environment. Thus, it is undesirable to keep the SCR system filled with ammonia at all times. However, during low temperatures when the SCR system is already operating at a lower efficiency, waiting for sufficient ammonia to adsorb in the SCR catalyst further decreases performance. Thus, low temperatures create challenges such as formation of solid deposits on the walls of the decomposition chamber and the SCR system not readily converting NOx.

To increase the efficiency of the decomposition chamber and the SCR system during low temperatures, the reductant may be inserted into the decomposition chamber in a vaporized form to reduce formation of solid deposits on the walls of the decomposition chamber. While the vaporizer is beneficial during the low temperatures, a vaporizer is not necessarily needed during normal temperature operation of the engine. In some cases, a special SCR catalyst that is optimized for low temperature operating conditions may be used in the SCR system. However, the optimized SCR catalyst by itself may not be enough to achieve the desired levels of NOx reduction. Keeping the SCR system filled with ammonia at all times is also not feasible since the ammonia tends to desorb at normal temperature operating conditions.

Thus, the present disclosure provides technical solutions for increasing the operating efficiency of the decomposition chamber and/or the SCR system. The aftertreatment system of the present disclosure provides a dual-leg system in which the exhaust gas exiting the engine may take a first exhaust gas path if the temperature of the exhaust gas is equal to or less than a predetermined temperature threshold and take a second exhaust gas path if the temperature of the exhaust gas is greater than the predetermined temperature threshold. Thus, in some embodiments, the exhaust gas may be directed to the first exhaust gas path during low temperatures and to the second exhaust gas path during normal temperatures. A selector valve may be used to divert the exhaust gas

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between the first exhaust gas path or the second exhaust gas path. The first exhaust gas path may be optimized for low temperature operation.

For example, in some embodiments, a heater may be used to heat at least a portion of the gas diverted to the first exhaust gas path. The heater may be activated for a period of time until the temperature of the exhaust gas attains a desired target temperature. In some embodiments, a controller may selectively and dynamically activate and deactivate the heater based on the current and desired temperatures in the decomposition chamber and/or within the SCR system using feedback inputs. In some embodiments, all of the exhaust gas may be heated during low temperature conditions, while in other embodiments, only a portion of the exhaust gas may be heated during the low temperature conditions. Heating even a portion of the exhaust gas may improve efficiency of the decomposition chamber and/or the SCR system during low temperatures.

Further, in some embodiments, a vaporizer may be used in the first exhaust gas path to further improve the efficiency of the decomposition chamber. In some embodiments, the SCR system may use an SCR catalyst optimized for low temperature operation. Because during low temperatures, the ammonia does not readily desorb, in some embodiments, the SCR system may be filled with ammonia at all times.

Thus, the present disclosure provides an effective mechanism for improving the efficiency of the aftertreatment system during low temperature operating conditions of the engine.

Referring now to FIG. 1, an example block diagram of an aftertreatment system 100 is shown, in accordance with some embodiments of the present disclosure. The aftertreatment system 100 is configured to receive exhaust gas from an engine 105. The engine 105 may be a compression ignited internal combustion engine such as a diesel engine, a spark-ignited internal combustion engine such as a gasoline engine, or any other types of engine such as a natural gas engine, a dual fuel engine, a biodiesel engine, an E-85 engine, etc. The engine 105 emits exhaust gas as a result of combustion of air from the atmosphere with fuel. The exhaust gas is discharged from the engine 105, via an inlet conduit 110, into a housing 115.

The housing 115 defines an internal volume within which one or more elements for treating the exhaust gas are disposed. To withstand the operating conditions, the housing 115 may be formed from a rigid, heat-resistant, and corrosion-resistant material such as stainless steel, iron, aluminum, metals, ceramics, or any other suitable material. Although the housing 115 has been shown in FIG. 1 as having a particular shape and size, the housing may have any suitable cross-section (e.g., circular, square, rectangular, oval, elliptical, polygonal, etc.) and any suitable size. The housing 115 may house an oxidation catalyst 120 for oxidizing nitric oxide and certain types of particulate matter from the exhaust gas, and decomposing unburnt hydrocarbons from the exhaust gas. In some embodiments, the oxidation catalyst 120 may be a diesel oxidation catalyst ("DOC") or other type of oxidation catalyst that is suitable for use in the aftertreatment system 100.

In some embodiments, the aftertreatment system 100 may include a hydrocarbon insertion assembly 125 for selectively injecting a hydrocarbon (e.g., fuel) into the oxidation catalyst 120. The oxidation catalyst 120 may catalyze ignition of the hydrocarbon so as to increase a temperature of the exhaust gas for regenerating the oxidation catalyst. In some embodiments, the aftertreatment system 100 may also include a particulate filter (not shown in FIG. 1) within the

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housing 115. The particulate filter may be disposed downstream or upstream of the oxidation catalyst 120. When positioned “upstream” of the oxidation catalyst 120, the particulate filter may be positioned between the inlet conduit 110 and the oxidation catalyst such that the exhaust gas exiting the particulate filter enters the oxidation catalyst. When positioned “downstream” of the oxidation catalyst 120, the particulate filter may be positioned between the oxidation catalyst and a selective catalytic reduction (“SCR”) system 130 such that the exhaust gas exiting the oxidation catalyst enters the particulate filter. In some embodiments, particulate filters may be provided both upstream and downstream of the oxidation catalyst 120.

The particulate filter may be configured to remove particulate matter (e.g., soot, debris, inorganic particles, etc.) from the exhaust gas. The particulate filter may be any of a variety of filters that are suitable for use within the after-treatment system 100. For example, in some embodiments, the particulate filter may be a diesel particulate filter (“DPF”) having a ceramic filter (e.g. cordierite), and may be symmetric or asymmetric. In some embodiments, the particulate filter may be catalyzed. In some embodiments, the oxidation catalyst 120 and the particulate filter may be integrated into a single component.

The housing 115 may also include the SCR system 130 that is configured to reduce constituents such as oxides of nitrogen (NO_x) gases, carbon monoxide (CO) gases, hydrocarbons, etc. from the exhaust gas. The SCR system 130 may include or be associated with a decomposition chamber that is configured to receive reductant from a reductant storage tank 135 via a reductant insertion assembly 140. A reductant port 145 may be positioned on a sidewall of the housing 115 to allow insertion of the reductant into an internal volume of the decomposition chamber. In some embodiments, the reductant port 145 may include a reductant injector that is configured to combine a flow of reductant received via the reductant port 145 with compressed air, and deliver a stream or a jet of the reductant-air combination into the decomposition chamber. In some embodiments, the reductant injector may be a nozzle of a predetermined diameter. In other embodiments, other mechanisms may be used to selectively deliver the reductant into the decomposition chamber. Thus, the decomposition chamber may be configured to receive exhaust gas and a reductant, and facilitate mixing of the exhaust gas with the reductant to form an exhaust gas-reductant mixture. In some embodiments, mixers, baffles, vanes, or other structures may be used in conjunction with the decomposition chamber to further facilitate mixing of the reductant with the exhaust gas.

The decomposition chamber may be positioned in a variety of locations. For example, in some embodiments, the decomposition chamber may be positioned upstream of the SCR system 130 to allow the reductant to be inserted, via the reductant port 145, into the decomposition chamber upstream of the SCR system. In other embodiments, the decomposition chamber may be structured such that the reductant port 145 is configured to allow the reductant to be inserted directly on the SCR system 130. In yet other embodiments, the decomposition chamber may be disposed in the inlet conduit 110. Thus, the positioning of the decomposition chamber may vary from one embodiment to another.

The reductant that is injected into the decomposition chamber via the reductant port 145 may be stored within the reductant storage tank 135. The reductant facilitates decomposition of the constituents of the exhaust gas (e.g., NO_x gases included in the exhaust gas). Any suitable reductant

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may be used depending upon the constituents of the exhaust gas. For example, in some embodiments, the exhaust gas may include a diesel exhaust gas and the reductant may include a diesel exhaust fluid (e.g., the diesel exhaust fluid marketed under the name ADBLUE®) such as urea, an aqueous solution of urea, or any other fluid that includes ammonia. When aqueous urea solution is used for the reductant, the solution may include a particular ratio of urea to water. For example, in some embodiments, the ratio may be 32.5% by volume of urea and 67.5% by volume of deionized water, 40% by volume of urea and 60% by volume of deionized water, or any other suitable ratio of urea to deionized water. The reductant from the reductant storage tank 135 may be selectively inserted into the decomposition chamber by the reductant insertion assembly 140. The reductant insertion assembly 140 may include various structures to facilitate receipt of the reductant from the reductant storage tank 135 and delivery to the reductant port 145. For example, the reductant insertion assembly 140 may include, or be associated with to control, various pumps, valves, screens, filters, etc. that facilitate receiving the reductant from the reductant storage tank 135 and supplying that reductant to the reductant port 145.

Further, in some embodiments, the reductant may be inserted into the decomposition chamber in liquid form or in gaseous form. In some embodiments, the reductant port 145, and particularly the reductant injector associated with the reductant port, may be configured to inject the reductant in liquid form. In such embodiments, the reductant port 145 may constitute or be part of a “wet doser” or “liquid doser.” In other embodiments, a vaporizer may be associated with the reductant port 145, the reductant injector, and/or the reductant insertion assembly 140 to vaporize or convert the liquid reductant into gaseous form before insertion into the housing 115. Whether inserted in liquid form or gaseous form, upon being injected into the decomposition chamber, the reductant undergoes evaporation, thermolysis, and/or hydrolysis to form gaseous ammonia, which is then mixed with the exhaust gas to form the exhaust gas-reductant mixture. The exhaust gas-reductant mixture may then flow over an SCR catalyst 150 of the SCR system 130.

The SCR catalyst 150 is formulated to decompose certain constituents in the exhaust gas using the gaseous ammonia as a reagent in the presence of the SCR catalyst. Specifically, the SCR catalyst 150 causes the gaseous ammonia in the exhaust gas to catalyze, thereby reducing the NO_x in the exhaust gas during the oxidation reaction. In some embodiments, the SCR catalyst 150 may include a metal-zeolite catalyst including, but not limited to Cu-CHA-Zeolite (e.g., a Cu-SSZ-13 catalyst), but also other zeolite structures including Cu-SAPO-34 catalyst, Cu-LTA, Cu-AEI, Cu-ZSM, Cu-beta, Cu-Chabazite, or any other suitable catalyst. In other embodiments, the SCR catalyst 150 may include a vanadium, an iron-zeolite, or a copper/iron-zeolite catalyst. In still other embodiments, the SCR catalyst 150 may include a multi-zone catalyst, for example, having a first zone including a copper-zeolite catalyst, and a second zone including an iron-zeolite catalyst, or vice versa. The SCR catalyst 150 may be disposed on a suitable substrate such as, for example, a ceramic (e.g., cordierite) or metallic (e.g., kanthal) monolith core which may, for example, define a honeycomb structure. A washcoat may also be used as a carrier material for the SCR catalyst 150. Such washcoat materials may include, for example, aluminum oxide, titanium dioxide, silicon dioxide, any other suitable washcoat material, or a combination thereof. The monolith core may be securely positioned in a can to form the SCR system 130,

which may be installed in the aftertreatment system **100**. In some embodiments, a heater **155** may be coupled to the SCR system **130** and configured to heat the exhaust gas within the SCR system and/or the decomposition chamber. In some embodiments, the SCR system **130** may include a selective catalytic reduction filter (SCRf). The treated exhaust gas (e.g., treated to reduce constituents such as NOx gases, unburnt hydrocarbons, etc.) is expelled, via an outlet conduit **160**, into the environment.

Although the oxidation catalyst **120**, the particulate filter, the decomposition chamber, and the SCR system **130** have been described as being disposed within a single housing (e.g., the housing **115**), in some embodiments, one or more of those components may be disposed in separate housings and connected together in operable association. Further, although a single instance of each of the oxidation catalyst **120**, the particulate filter, the decomposition chamber, and the SCR system **130** has been described, in some embodiments, multiple instances of one or more of those elements may be provided within the aftertreatment system **100**, if suitable.

Referring still to FIG. **1**, the aftertreatment system **100** also includes a controller **165** that is configured to control operation of the various elements of the aftertreatment system in treating the exhaust gas. For example, the controller **165** may be operably connected to the reductant insertion assembly **140** to instruct the reductant insertion assembly to selectively deliver the reductant from the reductant storage tank **135** to the reductant port **145**. The controller **165** may also be operably connected to the reductant port **145** to selectively operate the reductant port to insert the reductant received from the reductant storage tank **135** into the decomposition chamber. In some embodiments, the reductant insertion assembly **140** may be configured to control operation of the reductant port **145**.

The controller **165** may also be configured to control the hydrocarbon insertion assembly **125** to selectively insert hydrocarbons into the oxidation catalyst **120** and control the heater **155** to operate the heater when needed. The controller **165** may also be connected to other elements of the aftertreatment system **100** that are controlled by the controller. The controller **165** may be operably coupled to the various components of the aftertreatment system **100** using any type and any number of wired or wireless connections. For example, in some embodiments, a wired connection such as a serial cable, a fiber optic cable, a CAT5 cable, etc. may be used to communicably connect the controller **165** to one or more elements of the aftertreatment system **100**. In other embodiments, a wireless connection such as the Internet, Wi-Fi, cellular, radio, Bluetooth, ZigBee, etc. may be used. In some embodiments, a combination of wired and wireless connections may be used. Further, in some embodiments, a controller area network (CAN) bus may provide the exchange of signals, information, and/or data between the controller **165** and the various elements of the aftertreatment system **100**.

The controller **165** may include or be associated with one or more processing units. The processing unit(s) may include a microprocessor, programmable logic controller (PLC) chip, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a digital signal processor (DSP), a group of processing components, or other suitable electronic processing components. The processing unit(s) of the controller **165** may be configured to execute instructions for performing the operations described herein. The processing unit(s) may be implemented in hardware, firmware, software, or any com-

bination thereof. "Executing an instruction" means that the processing unit(s) may perform operation(s) called for by that instruction. The processing unit(s) may retrieve the instruction from a memory associated with the controller **165** for execution and copy the instruction in an executable form to a physical memory. In some embodiments, the processing unit(s) may be configured to execute the instruction without first copying the instruction to the physical memory. The instruction may be written using one or more programming languages, scripting languages, assembly languages, etc. Thus, the controller **165**, via its associated processing unit(s), may be configured to execute instructions, algorithms, commands, or programs stored in the memory associated with the controller.

Although a single controller (e.g., the controller **165**) configured to control multiple elements (e.g., the reductant insertion assembly **140**, the hydrocarbon insertion assembly **125**, the heater **155**, etc.) of the aftertreatment system **100** is shown, in some embodiments, separate controllers for one or more of those elements may be used. The controller **165** may include or be associated with other hardware, software, and/or firmware components that may be needed or considered useful to have in performing the functions described herein. The controller **165** may be configured to control the reductant insertion assembly **140**, the hydrocarbon insertion assembly **125**, the heater **155**, and any other element of the aftertreatment system **100** that is controlled by the controller based on data received from one or more sensors, such as sensors **170**, **175**, and **180**.

In some embodiments, the sensor **170** may be positioned to measure one or more parameters in the exhaust gas flowing through the inlet conduit **110**. The sensor **170** may include a NOx sensor configured to measure an amount of NOx gases in the exhaust gas flowing into the housing **115**. In some embodiments, the sensor **170** may include a temperature sensor to measure the temperature of the exhaust gas at the inlet of the housing **115**. In some embodiments, the sensor **170** may include a pressure sensor, an oxygen sensor, a particulate matter sensor, or any other sensor to measure a parameter that may be needed or considered desirable for the controller **165** to have in controlling the various elements of the aftertreatment system **100**. Further, although a single sensor (e.g., the sensor **170**) is shown in the inlet conduit **110**, in some embodiments, multiple sensors positioned at various locations of the inlet conduit may be used. Further, in some embodiments, a single instance of the sensor **170** may be configured to measure a single parameter (e.g., temperature, NOx amount, etc.), while in other embodiments, the single instance of the sensor may be configured to measure multiple parameters.

Similarly, the sensor **175** may be positioned to measure one or more parameters in the exhaust gas flowing through the outlet conduit **160**, and the sensor **180** may be positioned to measure one or more parameters associated with the engine **105**. Similar to the sensor **170**, the sensor **175** may include a single sensor or multiple sensors, and may be configured to measure needed or desirable parameter(s) such as amount of NOx gases expelled into the environment, temperature, pressure, particulate matter, ammonia (e.g., to determine ammonia slip), etc. Likewise, the sensor **180** may include a single sensor or multiple sensors, and may be configured to measure one or more operating parameters from the engine **105**.

Although the sensors **170**, **175**, and **180** have been shown in the aftertreatment system **100** as being positioned in specific locations, the positioning of those sensors may vary as suitable. Further, additional or fewer sensors across

various suitable locations of the aftertreatment system **100** may be used. The sensors **170**, **175**, **180** may be physical sensors or virtual sensors. The sensors **170**, **175**, **180** may be configured to collect data and transmit that data to the controller **165**. Based on the data, the controller **165** may then control operation of other elements (e.g., the hydrocarbon insertion assembly **125**, the reductant insertion assembly **140**, the heater **155**, etc.) of the aftertreatment system **100**.

Although certain components of the aftertreatment system **100** are shown and described in FIG. 1, the aftertreatment system may include other or additional elements that may be suitable. For example, in some embodiments, the aftertreatment system **100** may include an ammonia slip catalyst (“ASC”) or ammonia oxidation catalyst (“AMOX”) to reduce ammonia slip by which ammonia that is not catalyzed by the SCR catalyst **150** may be decomposed. In some embodiments, the aftertreatment system **100** may include mixers, baffle plates, secondary filters (e.g., a secondary partial flow or catalyzed filter), or any other component that may be needed or considered desirable in properly operating the aftertreatment system **100**.

Turning now to FIG. 2, an example block diagram of an aftertreatment system **200** is shown, in accordance with some embodiments of the present disclosure. The aftertreatment system **200** may include similar elements as the aftertreatment system **100**, although only some of those elements are shown in FIG. 2. The aftertreatment system **200** includes an engine **205** that emits exhaust gas into an inlet conduit **210**. A temperature sensor **215** is configured to measure the temperature of the exhaust gas flowing through the inlet conduit **210**. In some embodiments, the temperature sensor **215** may be a thermistor. In other embodiments, the temperature sensor **215** may be another type of temperature measuring device that is suitable for measuring the temperature of the exhaust gas flowing through the inlet conduit **210**. The exhaust gas from the inlet conduit **210** is directed through a DOC **220** for oxidizing hydrocarbons from the exhaust gas, and optionally, through a diesel particulate filter (“DPF”) **225** positioned downstream of the DOC for removing certain particulate matter from the exhaust gas. The DOC **220** is similar to the oxidation catalyst **120** and the DPF **225** is similar to the particulate filter discussed above.

From the DPF **225**, the exhaust gas is directed to a selector valve **230** positioned downstream of the DPF. The selector valve **230** is configured to divert the exhaust gas to a first exhaust gas path **235** or to a second exhaust gas path **240** based on the temperature of the exhaust gas as measured by the temperature sensor **215**. The first exhaust gas path **235** includes a first injector **245A** that injects a first reductant into a first decomposition chamber **250A** of the first exhaust path. The reductant in the first decomposition chamber **250A** is converted into gaseous ammonia, and mixed with exhaust gas that is diverted to the first exhaust gas path **235**. The mixture of the exhaust gas and gaseous ammonia is diverted to a first SCR **255A** positioned downstream of the first decomposition chamber **250A** in the first exhaust gas path **235**. In the first SCR **255A**, the gaseous ammonia is oxidized in the presence of an SCR catalyst to reduce NO_x gases in the exhaust gas. The treated exhaust gas is diverted to a first ASC **260A** that is positioned downstream of the first SCR **255A**. The first ASC **260A** decomposes any unreacted ammonia in the exhaust gas received from the first SCR **255A**. The exhaust gas from the first ASC **260A** is then expelled into the atmosphere via an outlet conduit **265**. The first exhaust gas path **235** also includes a heater **270** that is configured to heat the exhaust gas that has been diverted to

the first exhaust gas path. In some embodiments, the heater **270** may be an electric heater.

Similarly, the second exhaust gas path **240** includes a second injector **245B** for injecting a second reductant into a second decomposition chamber **250B** to generate gaseous ammonia therein. The exhaust gas that is diverted to the second exhaust gas path **240** flows into the second decomposition chamber **250B** and mixes with the gaseous ammonia. The mixture then enters a second SCR **255B** positioned downstream of the second decomposition chamber **250B**. In the second SCR **255B**, the gaseous ammonia is oxidized in the presence of an SCR catalyst to reduce NO_x in the exhaust gas. A second ASC **260B**, positioned downstream of the second SCR **255B**, decomposes any unreacted ammonia in the exhaust gas. The treated exhaust gas is expelled from the second exhaust gas path **240** into the environment via the outlet conduit **265**.

Thus, the exhaust gas exiting the DPF **225** may take one of two close-coupled parallel paths—the first exhaust gas path **235** or the second exhaust gas path **240**—based on a position of the selector valve **230**. The position of the selector valve **230** may be controlled by a controller **275** based on the temperature of the exhaust gas as measured by the temperature sensor **215**. The controller **275** is similar to the controller **165**. In some embodiments, the controller **275** may control the position of the selector valve **230** based on temperature of the exhaust gas measured at other locations such as outlet of the DOC **220** or the outlet of the DPF **225**.

In some embodiments, the selector valve **230** may be a multi position valve. In some embodiments, the selector valve **230** may, by default, be in a closed position that does not allow the exhaust gas to be diverted to either the first exhaust gas path **235** or the second exhaust gas path **240**. Upon receiving instructions from the controller **275** to divert the exhaust gas to the first exhaust gas path **235**, the selector valve **230** may move to a first open position that is configured to divert all of the exhaust gas to the first exhaust gas path. Similarly, upon receiving instructions from the controller **275** to divert the exhaust gas to the second exhaust gas path **240**, the selector valve **230** may move to a second open position that is configured to divert all of the exhaust gas to the second exhaust gas path. In some embodiments and as discussed further below, the position of the selector valve **230** may be adjustable between the first open position and the second open position such that portions of the exhaust gas may flow to both the first exhaust gas path **235** and the second exhaust gas path **240**. Therefore, the position of the selector valve **230** determines whether the exhaust gas is diverted to the first exhaust gas path **235**, the second exhaust gas path **240**, both the first and second exhaust gas paths, or neither.

The first exhaust gas path **235** may be configured for use during cold-start conditions when the temperature of the exhaust gas (e.g., as measured by the temperature sensor **215**) is below a predetermined temperature threshold. For example, in some embodiments, if the temperature of the exhaust gas is between about 70° C.-180° C., the first exhaust gas path **235** may be used for treating the exhaust gas exiting the DPF **225**. In some embodiments, the second exhaust gas path **240** may be used during normal conditions when the temperature of the exhaust gas is above 180° C. The temperature ranges for diverting the exhaust gas to either the first exhaust gas path **235** or the second exhaust gas path **240** may vary in other embodiments. Thus, the controller **275** may receive the temperature, as measured via the temperature sensor **215**, of the exhaust gas flowing through the inlet conduit **210**. Based on the temperature of

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the exhaust gas, the controller **275** may adjust the position of the selector valve **230** to divert all of the exhaust gas to either the first exhaust gas path **235** or the second exhaust gas path **240**. For example, if the predetermined temperature threshold is 180° C., the controller **275** may instruct the selector valve **230** to move to the first open position to divert all of the exhaust gas to the first exhaust gas path **235** when the temperature of the exhaust gas is equal to or less than 180° C. Similarly, if the temperature of the exhaust gas is above 180° C., the controller **275** may instruct the selector valve **230** to move to the second open position to divert all of the exhaust gas to the second exhaust gas path **240**.

Because the first exhaust gas path **235** is configured to be used during cold-start conditions, the first exhaust gas path **235** may be optimized for efficient operation during those cold-start conditions. For example, during cold-start conditions, the reductant that is injected into the first decomposition chamber **250A** in liquid form may be more susceptible to depositing on the walls of the first decomposition chamber, and decreasing the operating efficiency of the first decomposition chamber. Thus, in some embodiments, the first injector **245A** may be associated with a vaporizer to vaporize the reductant before injection. Injecting the reductant in vaporized form may reduce formation of reductant deposit on the walls of the first decomposition chamber **250A**. Reducing the formation of reductant deposit may increase the operating efficiency of the first decomposition chamber **250A**. In other embodiments, a commercial reductant delivery injection system or another mechanism that is configured to reduce reductant deposits in the first decomposition chamber **250A** during cold-start conditions may be used.

In some embodiments, instead of or in addition to using a vaporizer, the heater **270** may be used to heat the exhaust gas that has been diverted to the first exhaust gas path **235** to increase the temperature of the exhaust gas, which in turn may increase the temperature in the internal cavity of the first decomposition chamber **250A**. Providing sufficient heat to the first decomposition chamber **250A** may further assist in reducing the formation of reductant deposits on the walls of the first decomposition chamber **250A**. For example, in some embodiments, the heat from the heater **270** may reduce the droplet size of the reductant, whether in liquid or vaporized form, in the first decomposition chamber **250A**, thereby reducing the formation of reductant deposits in the first decomposition chamber.

The heater **270** may be controlled by the controller **275**. The controller **275** may activate the heater **270** based on the temperature of the exhaust gas that is diverted to the first exhaust gas path **235** and the target temperature that is desired in the first decomposition chamber **250A**. The controller **275** may be configured to deactivate the heater **270** when the target temperature within the first decomposition chamber **250A** is reached. A temperature sensor (not shown) may be positioned within the first decomposition chamber **250A** to measure the temperature in the internal cavity of the first decomposition chamber. The amount of time for which the heater **270** is activated may be based on the capacity of the heater and the amount of heat that is needed to attain the desired temperature. For example, in some embodiments, the heater may be configured such that 1 kilowatt of power of the heater **270** increases exhaust gas temperature at a rate of about 5° C. per second. In other embodiments, the heater **270** may be configured to attain other heating rates. Thus, the first exhaust gas path **235**, and particularly the first decomposition chamber **250A** of the first exhaust gas path, may be optimized for reducing reductant deposits by using

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a vaporizer to vaporize the reductant before insertion and/or by using heat from the heater **270** to vaporize the reductant (or reduce the droplet size of the reductant) after insertion. Other or additional mechanisms that may be configured to reduce reductant deposit in the first decomposition chamber **250A** may be used in other embodiments.

In some embodiments, the first SCR **255A** may also be optimized for operating during the cold-start conditions. For example, in some embodiments, the type of SCR catalyst that is used in the first SCR **255A** may be one that is more suitable for use during cold-start conditions. In some embodiments, the SCR catalyst in the first SCR **255A** may be a copper based, vanadium based, iron based, or a combination thereof. Further, in some embodiments, the first SCR **255A** may be configured to be filled with ammonia at all times. During normal operating conditions (e.g., when the temperature of the exhaust gas is above 180° C.), ammonia from an SCR catalyst may be desorbed and lost to the atmosphere. Thus, during normal operating conditions, continuous ammonia storage in the SCR catalyst is undesirable. However, during cold-start conditions, the ammonia that is adsorbed into the SCR catalyst is not readily desorbed.

Thus, in some embodiments, the SCR catalyst in the first SCR **255A** may be filled with ammonia at all times, such that the first SCR **255A** is able to reduce NOx from the exhaust gas as soon as the first SCR receives the mixture of gaseous ammonia and exhaust gas from the first decomposition chamber **250A** without having to first wait for the gaseous ammonia to adsorb in the SCR catalyst of the first SCR **255A**. In such embodiments in which the first SCR **255A** is configured for high ammonia storage at all times, the amount of reductant that is injected into the first decomposition chamber **250A** may be reduced to account for the ammonia storage in the first SCR. Thus, the SCR catalyst in the first SCR **255A** may be selected for high NOx reduction during low temperatures (e.g., during cold-start conditions) and high ammonia storage.

Further, the first SCR **255A** may be configured to start achieving a desired level of NOx reduction at a given temperature. The heater **270** may be used to heat the exhaust gas such that the first SCR **255A** reaches that given temperature in an internal cavity thereof. For example, in some embodiments, if the given temperature at which the first SCR **255A** starts achieving the desired level of NOx reduction is about 150° C., and the temperature of the exhaust gas entering the first exhaust gas path **235** (and/or at the inlet of the first SCR) is about 90° C., the controller **275** may activate the heater **270** until the temperature of the exhaust gas is at least 150° C. such that the heat from the exhaust gas heats the internal cavity of the first SCR **255A**. In some embodiments, the SCR catalyst within the first SCR **255A** may be a coupled reductant catalyst that may be configured to store the NOx species in the exhaust gas as ammonium nitrate below a certain temperature and release the NOx at a controlled rate when the first SCR heats up (e.g., using the heater **270**) to a certain temperature, thereby achieving optimal NOx reduction efficiency and accelerating soot oxidation across various filters under cold-start conditions. Thus, the first exhaust gas path **235**, and particularly the first SCR **255A** of the first exhaust gas path, may be optimized for achieving a desired level of NOx reduction by using an SCR catalyst that is optimized for cold-start conditions and/or optimized for high ammonia storage, and/or by using heat from the heater **270** to increase the temperature within the first SCR. Thus, the first decomposition chamber **250A**

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and/or the first SCR 255A may be configured for optimal operation during cold-start conditions.

Additionally, the controller 275 may be configured to dynamically control the insertion of reductant in the first decomposition chamber 250A based on one or more inputs. For example, in some embodiments, the controller 275 may be configured to control the amount of reductant that is inserted into the first decomposition chamber 250A based on the temperature of the exhaust gas entering the first exhaust gas path 235, the temperature within the cavity of the first SCR 255A, the total ammonia storage within the first SCR, ambient pressure, the desired NOx reduction efficiency, and/or the engine out NOx flux (e.g., amount of NOx in the exhaust gas exiting the outlet conduit 265). In other embodiments, the controller 275 may use other or additional inputs such as temperature of the exhaust gas in the outlet conduit 265, temperature of the exhaust gas at the inlet of the first SCR, amount of NOx at the inlet of the first SCR, etc., to dynamically vary the injection of reductant into the first decomposition chamber 250A. To dynamically vary the amount of reductant that is injected into the first decomposition chamber 250A, the controller may receive one or more of the above inputs and determine in real-time or substantial real-time the amount of reductant that needs to be inserted into the first decomposition chamber 250A. Upon determining the amount of reductant, the controller 275 may control the first injector 245A to insert the determined amount of the reductant. Although not shown, the first exhaust gas path 235 may include sensors installed at appropriate positions for providing data to the controller 275 based on which the controller may dynamically adjust the injection of the reductant in the first decomposition chamber 250A.

The controller 275 may also use data from one or more of such sensors to control the operation of the heater 270. For example, the controller 275 may determine the temperature within the internal cavity of the first decomposition chamber 250A and/or the first SCR 255A. The controller 275 may also know the target temperature at which the reductant deposits within the first decomposition chamber 250A are reduced. Thus, based on the temperature of the exhaust gas, the current temperature within the first decomposition chamber 250A, the target temperature within the first decomposition chamber, and the capacity (e.g., power) of the heater 270, the controller 275 may determine the temperature to which the exhaust gas needs to be heated and the amount of time for which the heater needs to be activated to achieve the target temperature within the first decomposition chamber. Similarly, based on the temperature of the exhaust gas, the temperature within the internal cavity of the first SCR 255A, the desired NOx reduction level, the temperature at which the desired NOx reduction level is achieved, and the capacity of the heater 270, the controller 275 may determine the temperature to which the exhaust gas needs to be heated and the time for which the heater needs to be activated for achieving the desired NOx reduction level.

Thus, by dynamically activating and deactivating the heater 270, and controlling the injection of reductant in the first decomposition chamber 250A in real-time (or substantially in real-time) based on feedback received during operation, the first exhaust gas path 235 provides a close-coupled system.

With respect to the second exhaust gas path 240, because the second exhaust gas path is configured for normal operation (e.g., when the temperature of the exhaust gas is above 180° C.), the second decomposition chamber 250B is not as susceptible to reductant deposits as the first decomposition chamber 250A. Thus, the reductant may continue to be

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injected in liquid form (e.g., using a wet or liquid doser) in the second decomposition chamber 250B, although if desired, a vaporizer may be associated with the second injector 245B to vaporize the reductant before injection into the second decomposition chamber. Similarly, a standard SCR may continue to be used for the second SCR 255B. A standard SCR may include an SCR catalyst that is not specifically designed for high ammonia storage and/or optimized for low temperatures.

Further, in some embodiments, the relative sizes of the first SCR 255A and the second SCR 255B may vary. For example, because the first SCR 255A is used for a shorter period of time (e.g., until the exhaust gas temperature becomes greater than 180° C.) compared to the second SCR 255B, the first SCR may be smaller in size than the second SCR. Similarly, in some embodiments, the first decomposition chamber 250A may be smaller in size than the second decomposition chamber 250B, and the amount of reductant that is injected into each of the first decomposition chamber and the second decomposition chamber may vary. In some embodiments, a fixed amount of reductant may be inserted into the second decomposition chamber 250B, while the controller 275 may dynamically adjust the amount of reductant that is inserted into the first decomposition chamber 250A. In other embodiments, the controller 275 may also dynamically adjust the amount of reductant that is inserted into the second decomposition chamber 250B.

Referring now to FIG. 3, an example block diagram of an aftertreatment system 300 is shown, in accordance with some embodiments of the present disclosure. The aftertreatment system 300 is similar to the aftertreatment system 100 in that the aftertreatment system 300 includes similar elements as the aftertreatment system 100, although only some of those elements are shown in FIG. 3. The aftertreatment system 300 includes an engine 305 from which exhaust gas enters into an inlet conduit 310. In the inlet conduit 310, the temperature of the exhaust gas is measured by a temperature sensor 315, which in some embodiments, may be a thermometer. From the inlet conduit 310, the exhaust gas passes into a DOC 320 and optionally through a DPF 325 positioned downstream of the DOC before being diverted by a selector valve 330 positioned downstream of the DPF. The DOC 320 is similar to the oxidation catalyst 120, the DPF 325 is similar to the particulate filter discussed above, and the selector valve 330 is similar to the selector valve 230. Thus, the selector valve 330 is a multi-position valve that diverts exhaust gas between a first exhaust gas path 335 and a second exhaust gas path 340 based on instructions received from a controller 345. The controller 345 is similar to the controller 165.

Similar to the first exhaust gas path 235, the first exhaust gas path 335 is configured for use during cold-start conditions (e.g., when the temperature of the exhaust gas is between about 70° C.-180° C.). The first exhaust gas path 335 includes a first injector 350A for injecting reductant into a first decomposition chamber 355A, while the second exhaust gas path 340 includes a second injector 350B to inject reductant into a second decomposition chamber 355B. The first exhaust gas path 335 and the second exhaust gas path 340 combine to form a combined exhaust gas path 360. The combined exhaust gas path 360 is downstream of the first decomposition chamber 355A and the second decomposition chamber 355B. The combined exhaust gas path 360 includes an SCR 365 and an ASC 370 downstream of the SCR. The exhaust gas from the ASC 370 is emitted into the atmosphere using an outlet conduit 375.

Thus, in contrast to the aftertreatment system **200** in which each of the first exhaust gas path **235** and the second exhaust gas path **240** has their respective SCR (e.g., the first SCR **255A**, the second SCR **255B**) and ASC (e.g., the first ASC **260A**, the second ASC **260B**), the first exhaust gas path **335** and the second exhaust gas path **340** do not include SCR and ASC. Rather, the mixture of the exhaust gas and the gaseous ammonia from each of the first exhaust gas path **335** and the second exhaust gas path **340** flows into the SCR **365** of the combined exhaust gas path **360**. The SCR **365** oxidizes the ammonia in the presence of an SCR catalyst for reducing NO_x in the exhaust gas and the ASC **370** decomposes any unreacted ammonia from the exhaust gas. In some embodiments, the SCR **365** may be a standard SCR.

Further, the selector valve **330** may be configured to divert at least a portion of the exhaust gas to the first exhaust gas path **335** when the temperature (e.g., as measured by the temperature sensor **315**) of the exhaust gas is equal to or below a predetermined temperature threshold (e.g., 180° C.). Specifically, the controller **345** may determine the temperature of the exhaust gas flowing through the inlet conduit **310** and adjust the position of the selector valve **330** to a first open position or to a position between the first open position and the second open position to divert at least a portion the exhaust gas to the first exhaust gas path **335** if the temperature of the exhaust gas is less than or equal to the predetermined temperature threshold (e.g., 180° C.).

In some embodiments, the controller **345** may be configured to divert all of the exhaust gas to the first exhaust gas path **335** when the temperature of the exhaust gas is equal to or below predetermined temperature threshold (e.g., 180° C.). In such embodiments, the controller **345** may adjust the position of the selector valve **330** to the first open position. In other embodiments, the controller **345** may be configured to divert only a portion of the exhaust gas to the first exhaust gas path **335** when the temperature of the exhaust gas is equal to or below the predetermined temperature threshold (e.g., 180° C.). In such embodiments, the controller **345** may adjust the position of the selector valve **330** to be in between the first open position and the second open position based on the portion of the exhaust gas that is to be diverted to the first exhaust gas path **335**. Further, any exhaust gas that is not diverted to the first exhaust gas path **335** is diverted to the second exhaust gas path **340**.

The portion of the exhaust gas that is diverted to the first exhaust gas path **335** may be pre-determined. For example, in some embodiments, about 50% of the exhaust gas may be diverted to the first exhaust gas path **335** when the temperature of the exhaust gas is equal to or below the predetermined temperature threshold (e.g., 180° C.). The remaining about 50% of the exhaust gas may be diverted to the second exhaust gas path **340**. To divert about 50% of the exhaust gas to the first exhaust gas path **335**, the controller **345** may adjust the position of the selector valve **330** to be somewhat mid-way between the first open position and the second open position. In other embodiments, a different proportion of the exhaust gas may be diverted to the first exhaust gas path **335**. Thus, by varying the position of the selector valve **330** between the first open position and the second open position, the controller **345** may divert a portion of the exhaust gas to the first exhaust gas path **335** and a portion of the exhaust gas to the second exhaust gas path **340** when the temperature of the exhaust gas is less than or equal to the predetermined temperature threshold (e.g., 180° C.). When the temperature of the exhaust gas is above the predeter-

mined temperature threshold (e.g., 180° C.), the controller **345** may divert all of the exhaust gas to the second exhaust gas path **340**.

Further, although the temperature of the exhaust gas based on which the controller **345** controls the selector valve **330** is measured in the inlet conduit **310**, in some embodiments, the temperature of the exhaust gas may be measured at the outlet of the DOC **320** and/or the outlet of the DPF **325**.

Additionally, the exhaust gas that is diverted to the first exhaust gas path **335** may be heated by a heater **380**. The heater **380** may be controlled by the controller **345** to reduce reductant deposits in the first decomposition chamber **355A**. Thus, based on the temperature of the exhaust gas entering the first exhaust gas path **335**, the temperature of the exhaust gas in the first decomposition chamber **355A**, the desired temperature within the first decomposition chamber, the capacity of the heater **380**, and any other inputs that may be useful in increasing the efficiency of the first decomposition chamber, the controller may activate the heater for a period of time to heat the internal cavity of the first decomposition chamber. In some embodiments, the first decomposition chamber **355A** may also be configured to receive the reductant in vaporized form using a vaporizer coupled with the first injector **350A** to reduce reductant deposits on the walls of the first decomposition chamber. Similarly, in some embodiments, the reductant may be injected in the second decomposition chamber **355B** in liquid form using a wet or liquid doser. In some embodiments, the second injector **350B** may be coupled to a vaporizer to vaporize the reductant before insertion into the second decomposition chamber **355B**.

Additionally, in some embodiments and as discussed above, the controller **345** may be configured to dynamically adjust the amount of reductant that is inserted into the first decomposition chamber **355A** and/or the second decomposition chamber **355B**. For example, the controller **345** may receive data in real-time or substantial real-time from one or more components of the first exhaust gas path **335** and the combined exhaust gas path **360** to dynamically vary the amount of reductant being inserted into the first decomposition chamber **355A**. Similarly, the controller **345** may receive data from one or more components of the second exhaust gas path **340** and the combined exhaust gas path **360** to dynamically vary the amount of reductant being inserted into the second decomposition chamber **355B**. Thus, the first exhaust gas path **335** along with the combined exhaust gas path **360** forms a first parallel close-coupled system, while the second exhaust gas path **340** along with the combined exhaust gas path forms another parallel close-coupled system.

Referring to FIGS. **4A** and **4B**, example block diagrams of aftertreatment systems **400** and **400'**, respectively, are shown, in accordance with some embodiments of the present disclosure. The aftertreatment systems **400** and **400'** are similar to the aftertreatment system **100** in that the aftertreatment systems **400** and **400'** include similar elements as the aftertreatment system **100**, although only some of those elements are shown in FIGS. **4A** and **4B**. The aftertreatment system **400** shows a first serial implementation, while the aftertreatment system **400'** shows a second serial implementation.

Referring specifically to FIG. **4A**, the aftertreatment system **400** includes an engine **405** from which exhaust gas enters into an inlet conduit **410**. In the inlet conduit **410**, the temperature of the exhaust gas is measured by a temperature sensor **415**, which in some embodiments, may be a thermistor. From the inlet conduit **410**, the exhaust gas is diverted

by a selector valve 420. The selector valve 420 is similar to the selector valve 230. The selector valve 420 diverts the exhaust gas from the inlet conduit 410 to either a first exhaust gas path 425 or a second exhaust gas path 430. The first exhaust gas path 425 and the second exhaust gas path 430 combine together to form a combined exhaust gas path 435 downstream of the first exhaust gas path and the second exhaust gas path.

The first exhaust gas path 425 may be used during cold-start conditions to heat the exhaust gas to a desired temperature by a heater 440 before entering the combined exhaust gas path 435. Thus, when the temperature of the exhaust gas is at or below a predetermined temperature threshold (e.g., 180° C.), a controller 445 may adjust the position of the selector valve 420 to the first open position to divert all of the exhaust gas into the first exhaust gas path 425. When the temperature of the exhaust gas is above the predetermined temperature threshold (e.g., 180° C.), the controller 445 may adjust the position of the selector valve 420 to the second open position to divert all of the exhaust gas to the second exhaust gas path 430. In the first exhaust gas path 425, the exhaust gas is heated by the heater 440 to a desired temperature. The temperature to which the exhaust gas is heated may be dependent upon the target temperature that is desired in one or more elements of the combined exhaust gas path 435. In some embodiments, a heater may also be provided in the second exhaust gas path 430 and/or in the combined exhaust gas path 435.

The heated exhaust gas from the first exhaust gas path 425 enters the combined exhaust gas path 435, particularly, a DOC 450 of the combined exhaust gas path. The exhaust gas may optionally pass from the DOC 450 to a DPF 455 positioned downstream of the DOC. The DOC 450 is similar to the oxidation catalyst 120 and the DPF 455 is similar to the particulate filter discussed above with respect to FIG. 1. From the DPF 455, the heated exhaust gas is mixed with reductant in a first decomposition chamber 460. The reductant is inserted into the first decomposition chamber 460 by a first injector 465. The reductant may be inserted in liquid form (e.g., using a wet or liquid doser) or in vaporized form (e.g., using a vaporizer). The mixture of the exhaust gas and the reductant is diverted through a first SCR 470 and an ASC 475 of the combined exhaust gas path 435 before being released into the atmosphere via an outlet conduit 480. The first SCR 470 may be a standard SCR.

The controller 445 may be configured to heat the exhaust gas in the first exhaust gas path 425 to achieve a target temperature within the first decomposition chamber 460 and/or the first SCR 470. By positioning the first decomposition chamber 460 and the first SCR 470 in the combined exhaust gas path 435, the same decomposition chamber and SCR may be used for both the first exhaust gas path 425 and the second exhaust gas path 430. Further, by using the heater 440 to heat the exhaust gas diverted to the first exhaust gas path 425 to achieve a target temperature within the first decomposition chamber 460 and/or the first SCR 470, the operating efficiency of the first decomposition chamber and/or the first SCR may be increased during cold-start conditions.

When the controller 445 diverts the exhaust gas to the second exhaust gas path 430, the exhaust gas enters a second decomposition chamber 485A that receives reductant via a second injector 485B. The second decomposition chamber 485A may be configured to receive the reductant in either liquid form or vaporized form. In some embodiments, the mixing of the exhaust gas and the gaseous ammonia from the reductant may be facilitated by using a mixer 490 positioned

downstream of the second decomposition chamber 485A. Although not shown, a mixer may be used in conjunction with the first and second decomposition chambers in the embodiments of FIGS. 2 and 3 as well. The mixture of the exhaust gas and the reductant from the mixer 490 is diverted to a second SCR 495 before passing the exhaust gas to the combined exhaust gas path 435. In the combined exhaust gas path 435, the exhaust gas passes through the first decomposition chamber 460, the first SCR 470, and the ASC 475 before being emitted via the outlet conduit 480.

Thus, the exhaust gas from the inlet conduit 410 passes through either the first exhaust gas path 425 or the second exhaust gas path 430 before passing through the combined exhaust gas path 435. Further, in some embodiments, the controller 445 may be configured to dynamically adjust the amount of reductant being inserted into the first decomposition chamber 460 and/or the second decomposition chamber 485A based on feedback received from one or more components of the aftertreatment system 400. For example, when the exhaust gas is being diverted through the first exhaust gas path 425, the controller 445 may dynamically adjust the amount of reductant being inserted into the first decomposition chamber 460 based on feedback from one or more components of the first exhaust gas path 425 and the combined exhaust gas path 435. When the exhaust gas is being diverted through the second exhaust gas path 430, the controller 445 may dynamically adjust the amount of reductant being inserted into the first decomposition chamber 460 and/or the second decomposition chamber 485A based on feedback from one or more components of the second exhaust gas path and the combined exhaust gas path 435.

The embodiment of FIG. 4B is largely similar to the embodiment of FIG. 4A except that the first exhaust gas path and the second exhaust gas path are reversed in FIG. 4B. Thus, the aftertreatment system 400' includes an engine 405' from which exhaust gas enters an inlet conduit 410'. The temperature of the exhaust gas may be measured in the inlet conduit 410' by a temperature sensor 415', which in some embodiments may be a thermistor. A selector valve 420' may divert the exhaust gas to either a first exhaust gas path 425' or a second exhaust gas path 430'. The first exhaust gas path 425' and the second exhaust gas path 430' may combine together to form a combined exhaust gas path 435' downstream of the first exhaust gas path and the second exhaust gas path. When a controller 440' determines that the temperature of the exhaust gas as measured by the temperature sensor 415' is equal to or less than a predetermined temperature threshold (e.g., 180° C.), the controller may adjust a position of the selector valve 420' to the first open position to divert all of the exhaust gas to the first exhaust gas path 425'. When the controller 440' determines that the temperature of the exhaust gas is greater than the predetermined temperature threshold (e.g., 180° C.), the controller may adjust the position of the selector valve 420' to the second open position to divert all of the exhaust gas to the second exhaust gas path 430'. From the first exhaust gas path 425' or the second exhaust gas path 430', the exhaust gas enters the combined exhaust gas path 435'.

The combined exhaust gas path 435' may include a DOC 445', a DPF 450' downstream of the DOC, a first decomposition chamber 455' downstream of the DPF, an injector 460' for injecting reductant into the first decomposition chamber, a first SCR 465' downstream of the first decomposition chamber, and an ASC 470' downstream of the first SCR. The exhaust gas is emitted into the atmosphere via an outlet conduit 475' downstream of the ASC 470'.

The first exhaust gas path **425'** includes a heater **480'** configured to heat the exhaust gas diverted to the first exhaust gas path by the selector valve **420'**. The heated exhaust gas is mixed with gaseous ammonia within a second decomposition chamber **485A'** that receives reductant via a second injector **485B'**. A mixer **490'** may facilitate the mixing of the exhaust gas with the gaseous ammonia. A second SCR **495'** then oxidizes at least a portion of the gaseous ammonia to reduce NOx in the exhaust gas before flowing into the combined exhaust gas path **435'** to be further treated. The controller **440'** may heat the exhaust gas to a temperature that is desired in the second decomposition chamber **485A'**, the second SCR **495'**, the first decomposition chamber **455'**, and/or the first SCR **465'**. Further, as discussed above, the controller **440'** may dynamically adjust the amount of reductant that is inserted into the first decomposition chamber **455'** and/or the second decomposition chamber **485A'** based on feedback from one or more inputs. In some embodiments, a heater may also be provided in the second exhaust gas path **430'** and/or the combined exhaust gas path **435'**.

Turning to FIG. 5, an example flow chart outlining a process **500** for treating exhaust gas using an aftertreatment system is shown, in accordance with some embodiments of the present disclosure. The process **500** may be implemented using the aftertreatment systems **200**, **300**, **400**, **400'** above, and particularly, by the controllers (e.g., the controller **275**, the controller **345**, the controller **445**, the controller **440'**) of those aftertreatment systems. The process starts at operation **505** with the engine (e.g., the engine **205**, the engine **305**, the engine **405**, the engine **405'**) emitting the exhaust gas into the inlet conduit (e.g., the inlet conduit **210**, the inlet conduit **310**, the inlet conduit **410**, the inlet conduit **410'**). At operation **510**, the controller receives the temperature of the exhaust gas in the inlet conduit. For example, the controller **275** may receive the temperature of the exhaust gas in the inlet conduit **210** as measured by the temperature sensor **215**. Similarly, the controller **345** may receive the temperature of the exhaust gas as measured by the temperature sensor **315** in the inlet conduit **310**, while the controller **445** may receive the temperature of the exhaust gas as measured by the temperature sensor **415** in the inlet conduit **410**, and the controller **440'** may receive the temperature of the exhaust gas as measured via the temperature sensor **415'** in the inlet conduit **410'**.

In other embodiments, the controller **275** may additionally or alternatively determine the temperature of the exhaust gas at other locations such as the outlet of the DPF **225** and/or the outlet of the DOC **220**. Similarly, in some embodiments, the controller **345** may additionally or alternatively determine the temperature of the exhaust gas at the outlet of the DOC **320** and/or the outlet of the DPF **325**. When the controller determines the temperature of the exhaust gas at multiple locations, the controller may apply a mathematical function (e.g., average) to the various determined exhaust gas temperatures and adjust the selector valve (e.g., the selector valve **230**, the selector valve **330**, the selector valve **420**, the selector valve **420'**) based on the computed average temperature.

At operation **515**, the controller determines whether the temperature of the exhaust gas as determined at the operation **510** is less than a predetermined temperature threshold (e.g., 180° C.). If the controller determines that the temperature of the exhaust gas is greater than the predetermined threshold, the controller adjusts the position of the selector valve at operation **520** to the second open position to divert all of the exhaust gas to the second exhaust gas path (e.g.,

the second exhaust gas path **240**, the second exhaust gas path **340**, the second exhaust gas path **430**, the second exhaust gas path **430'**). Alternatively, if at the operation **515**, the controller determines that the temperature of the exhaust gas is equal to or less than the predetermined temperature threshold, the controller adjusts the position of the selector valve at operation **525** to the first open position or a position between the first open position and the second open position to divert at least a portion of the exhaust gas to the first exhaust gas path (e.g., the first exhaust gas path **235**, the first exhaust gas path **335**, the first exhaust gas path **425**, the first exhaust gas path **425'**), as discussed above.

For example, in the aftertreatment systems **200**, **400**, **400'**, upon determining that the temperature of the exhaust gas is equal to or below the predetermined temperature threshold, the controller **275**, the controller **445**, and the controller **440'** respectively, divert all of the exhaust gas to the respective first exhaust gas path **235**, the first exhaust gas path **425**, and the first exhaust gas path **425'**. In contrast, in the aftertreatment system **300**, upon determining that the temperature of the exhaust gas the controller is equal to or below the predetermined temperature threshold, the controller **345** diverts only a predetermined portion of the exhaust gas to the first exhaust gas path **335**. The controller **345** diverts the remaining portion of the exhaust gas to the second exhaust gas path **340**.

In some embodiments, the controller may rely upon inputs in addition to the temperature of the exhaust gas in controlling the selector valve. For example, in some embodiments, the controller may receive inputs from an NOx sensor positioned at the inlet and/or outlet of the DOC (e.g., the DOC **220**, the DOC **320**, the DOC **450**, the DOC **445'**), inlet and/or outlet of the DPF (e.g., the DPF **225**, the DPF **325**, the DPF **450'**, the DPF **455**), and/or at the outlet conduit (e.g., the outlet conduit **265**, the outlet conduit **375**, the outlet conduit **475'**, the outlet conduit **480**) to determine an ammonia to NOx ratio (ANR) of the exhaust gas. In other embodiments, the controller may receive inputs from NOx sensors positioned at other or additional locations to determine the ANR. Based upon the ANR and the temperature of the exhaust gas, the controller may control the selector valve to divert the exhaust gas between the first exhaust gas path and/or the second exhaust gas path to achieve a desired NOx reduction efficiency. In some embodiments, the controller may rely only on ANR to control the selector valve. In other embodiments, the controller may use other or additional inputs to control the selector valve.

Further, at operation **530**, when the exhaust gas is diverted to the first exhaust gas path, the controller activates a heater (e.g., the heater **270**, the heater **380**, the heater **440**, or the heater **480'**) to heat the exhaust gas diverted to the first exhaust gas path. As discussed above, the controller may activate the heater to heat the exhaust gas until the temperature of the exhaust gas attains a desired temperature. The desired temperature may be based on a temperature in the decomposition chamber and/or the SCR that is desired in which the exhaust gas is to flow. The controller may receive feedback data on the current operating conditions in the decomposition chamber, the SCR system, and/or other components to determine the desired temperature. Upon determining that the desired temperature has been attained, the controller deactivates the heater at operation **535**. The process **500** ends at operation **540**.

It should be noted that the term "example" as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term

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is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

As used herein, the term “about” generally mean plus or minus 10% of the stated value. For example, about 0.5 would include 0.45 and 0.55, about 10 would include 9 to 11, about 1000 would include 900 to 1100.

The term “coupled” and the like as used herein mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

It is important to note that the construction and arrangement of the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements; values of parameters, mounting arrangements; use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. Additionally, it should be understood that features from one embodiment disclosed herein may be combined with features of other embodiments disclosed herein as one of ordinary skill in the art would understand. Other substitutions, modifications, changes, and omissions may also be made in the design, operating conditions, and arrangement of the various exemplary embodiments without departing from the scope of the present embodiments.

While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any embodiments or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular embodiments. Certain features described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a sub combination.

What is claimed is:

1. An aftertreatment system comprising:

a first exhaust gas path comprising a heater;
a second exhaust gas path comprising a first decomposition chamber configured to receive reductant and a first selective catalytic reduction catalyst downstream of the first decomposition chamber;

a combined exhaust gas path downstream of the first exhaust gas path and the second exhaust gas path, the combined exhaust gas path configured to receive exhaust gas from both the first exhaust gas path and the second exhaust gas path;

a selector valve configured to divert the exhaust gas between the first exhaust gas path and the second exhaust gas path based on a temperature of the exhaust gas; and

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a controller programmed to control the selector valve such that:

the selector valve diverts the exhaust gas to the first exhaust gas path when the temperature of the exhaust gas is equal to or less than a predetermined temperature threshold; and

the selector valve diverts the exhaust gas to the second exhaust gas path when the temperature of the exhaust gas is greater than the predetermined temperature threshold; wherein:

the heater is configured to heat the exhaust gas received in the first exhaust gas path;

the first decomposition chamber is configured to receive the exhaust gas that has been diverted to the second exhaust gas path by the selector valve; and

the first selective catalytic reduction catalyst is configured to receive the exhaust gas from the first decomposition chamber.

2. The aftertreatment system of claim 1, wherein:

the combined exhaust gas path further comprises:

a second decomposition chamber to receive the exhaust gas from the first exhaust gas path and the second exhaust gas path, and

a second selective catalytic reduction catalyst downstream of the second decomposition chamber to receive the exhaust gas from the second decomposition chamber.

3. The aftertreatment system of claim 2, wherein the combined exhaust gas path further comprises an ammonia slip catalyst downstream of the second selective catalytic reduction catalyst.

4. The aftertreatment system of claim 2, wherein the combined exhaust gas path further comprises an oxidation catalyst upstream of the second decomposition chamber.

5. The aftertreatment system of claim 4, wherein the combined exhaust gas path further comprises a particulate filter downstream of the oxidation catalyst and upstream of the second decomposition chamber.

6. The aftertreatment system of claim 2, wherein each of the first decomposition chamber and the second decomposition chamber is configured to receive reductant in either liquid form or vaporized form.

7. The aftertreatment system of claim 2, wherein the controller is further programmed to dynamically adjust an amount of reductant inserted into the first decomposition chamber and the second decomposition chamber based on feedback from one or more components of the aftertreatment system.

8. The aftertreatment system of claim 2, wherein each of the first selective catalytic reduction catalyst and the second selective catalytic reduction catalyst comprises a copper based catalyst, an iron based catalyst, or a vanadium based catalyst.

9. The aftertreatment system of claim 1, wherein the first selective catalytic reduction catalyst comprises a mixer to mix the exhaust gas with reductant.

10. The aftertreatment system of claim 1, wherein the predetermined temperature threshold is in a range of 70° C. to 180° C.

11. A method comprising:

determining, by a controller associated with an aftertreatment system, a temperature of exhaust gas;

comparing, by the controller, the temperature of the exhaust gas with a predetermined temperature threshold;

when the temperature of the exhaust gas is equal to or less than the predetermined temperature threshold, adjust-

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ing a selector valve to a first position so as to divert the exhaust gas to a first exhaust gas path, and heating the exhaust gas in the first exhaust gas path; and
 when the temperature of the exhaust gas is greater than the predetermined temperature threshold, adjusting the selector valve to a second position so as to divert the exhaust gas to a second exhaust gas path; wherein:
 a first decomposition chamber of the second exhaust gas path is configured to receive the exhaust gas that has been diverted to the second exhaust gas path by the selector valve;
 a first selective catalytic reduction catalyst of the second exhaust gas path is configured to receive the exhaust gas from the first decomposition chamber; and
 the exhaust gas from the first exhaust gas path and the second exhaust gas path is diverted to a combined exhaust gas path that is downstream of the first exhaust gas path and the second exhaust gas path and configured to receive the exhaust gas from the first exhaust gas path and the second exhaust gas path.

12. The method of claim 11, further comprising dynamically adjusting, by the controller, an amount of reductant inserted into the first decomposition chamber based on feedback from one or more components of the aftertreatment system.

13. The method of claim 12, further comprising injecting, by the controller, the amount of reductant in the first decomposition chamber in either liquid form or vaporized form.

14. The method of claim 11, further comprising dynamically adjusting, by the controller, an amount of reductant inserted into a second decomposition chamber based on feedback from one or more components of the aftertreatment system, wherein the second decomposition chamber is located in the combined exhaust gas path upstream of a second selective catalytic reduction catalyst that receives the exhaust gas from the second decomposition chamber.

15. The method of claim 11, wherein the predetermined temperature threshold is in a range of 70° C. to 180° C.

16. A non-transitory computer-readable media comprising computer readable instructions stored thereon that when executed by a processor of an aftertreatment system causes the processor to:

- determine a temperature of exhaust gas;
- compare the temperature of the exhaust gas with a predetermined temperature threshold;

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when the temperature of the exhaust gas is equal to or less than the predetermined temperature threshold, adjust a selector valve to a first position so as to divert the exhaust gas to a first exhaust gas path, and heat the exhaust gas in the first exhaust gas path; and
 when the temperature of the exhaust gas is greater than the predetermined temperature threshold, adjust the selector valve to a second position so as to divert the exhaust gas to a second exhaust gas path; wherein:
 a first decomposition chamber of the second exhaust gas path is configured to receive the exhaust gas that has been diverted to the second exhaust gas path by the selector valve;
 a first selective catalytic reduction catalyst of the second exhaust gas path is configured to receive the exhaust gas from the first decomposition chamber; and
 the exhaust gas from the first exhaust gas path and the second exhaust gas path is diverted to a combined exhaust gas path that is downstream of the first exhaust gas path and the second exhaust gas path and configured to receive the exhaust gas from the first exhaust gas path and the second exhaust gas path.

17. The non-transitory computer-readable media of claim 16, wherein the processor further executes computer-readable instructions to dynamically adjust an amount of reductant inserted into the first decomposition chamber based on feedback from one or more components of the aftertreatment system.

18. The non-transitory computer-readable media of claim 17, wherein the processor further executes computer-readable instructions to inject the amount of reductant in the first decomposition chamber in either liquid form or vaporized form.

19. The non-transitory computer-readable media of claim 16, wherein the processor further executes computer-readable instructions to dynamically adjust an amount of reductant inserted into a second decomposition chamber based on feedback from one or more components of the aftertreatment system, and wherein the second decomposition chamber is located in the combined exhaust gas path upstream of a second selective catalytic reduction catalyst that receives the exhaust gas from the second decomposition chamber.

20. The non-transitory computer-readable media of claim 16, wherein the predetermined temperature threshold is in a range of 70° C. to 180° C.

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